8.5 Trapezoids

Lesson Introduction

Benchmarks	MA.912.GR.1.5 Prove relationships and theorems about trapezoids. Solve mathematical and real-world problems involving postulates, relationships, and theorems of trapezoids. MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.	Learning Targets	• I can prove relationships and theorems about trapezoids. • I can use relationships and theorems about trapezoids.

Benchmark Coherence	Building On	Working Toward
	N/A	N/A

Essential Questions	• What are the properties of a trapezoid? • What are the properties of an isosceles trapezoid? • How can I prove relationships and theorems about trapezoids, and use them to solve problems?
Lesson Narrative	This lesson introduces students to another subset of quadrilaterals called trapezoids. Students will learn the parts of a trapezoid (bases, base angles, and legs) and explore features of trapezoids to write conjectures. Students will apply their knowledge of trapezoids to prove relationships and theorems about trapezoids and use relationships to solve problems.

Component Summary				
	8.5.1 Warm-Up (~5 min.)	Identify and define trapezoids and trapeziums. *(SE p. 33)*	8.5.4 Guided Instruction (10-15 min.)	Prove theorems about trapezoids. *(SE p. 36)*
	8.5.2 Collaboration (10 min.)	Determine the properties of isosceles trapezoids. *(SE p. 33)*	8.5.5 Guided Practice (10 min.)	Solve problems using the properties of trapezoids. *(SE p. 38)*
	8.5.3 Exploration (5-10 min.)	Name the parts of trapezoids. *(SE p. 35)*	8.5.6 Wrap-Up (~5 min.)	Solve problems using the properties of trapezoids. *(SE p. 38)*

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms:</u> • Trapezoid • Isosceles Trapezoid • Median • Midsegment • Altitude <u>Additional Terms:</u> • Base Angles • Bases • Legs	• Patty Paper • Rulers • Protractors	• N/A

	Common Misconceptions	Things to Consider
Teacher Tips	• Students may use the wrong base angles for 8.5.4 Guided Instruction, question 3.	• Consider creating anchor charts of the features of trapezoids and isosceles trapezoids.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Three Reads Consider the Three Reads routine for the 8.5.1 Warm-Up. This routine supports reading comprehension of problems and meta-awareness of mathematical language. It also supports negotiating information in a text with a partner in mathematical conversation.	**MA.K12.MTR.4.1: Discussion and Reflection** As students progress through 8.5.2 Collaboration, they are challenged to evaluate a variety of statements. Once the statements are determined to be incorrect, they will discuss writing a new conjecture with their partner. Students will also have the opportunity to compare and discuss their conjectures with another pair of partners.
Differentiate Instruction	For 8.5.4 Guided Instruction, consider implementing a silent board game. Display the proof, and have students volunteer to go to the board and fill in one of the missing statements or reasons. There is no student talk during this activity. This routine offers students entry into the problem where they feel confident.	

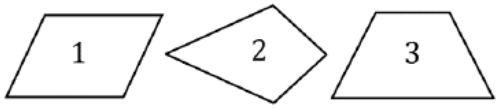
Lesson Notes:

8.5.1 Warm-Up: Trapezoid or Trapezium?

Component Overview: *Students review the American and British definitions of Trapezoid and Trapezium, and identify figures that fit each definition.*

8.5.1 Warm-Up: Trapezoid or Trapezium?

1. Which quadrilateral is classified as a trapezoid?



There is more than one definition of a trapezoid.

Inclusive Definition	Exclusive Definition
A quadrilateral with at least one pair of parallel opposite sides.	A quadrilateral with exactly one pair of parallel opposite sides.

There is more than one definition of trapezium.

American Definition	British Definition
A quadrilateral with no parallel sides.	A quadrilateral with two sides parallel.

2. Use the definitions to classify quadrilaterals 1, 2, and 3. Write the numbers of the quadrilateral in the table.

Inclusive Definition of Trapezoid	Exclusive Definition of Trapezoid
1, 3	3
American Definition of Trapezium	**British Definition of Trapezium**
2	1, 3

Three Reads
These definitions have minute differences that can be easily missed. Consider implementing this routine to ensure students notice these differences:

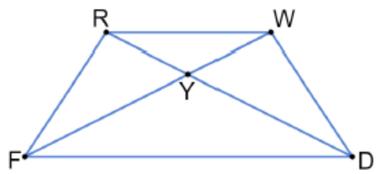
- *Students read through the definitions silently.*
- *Students read the definitions again silently, and make notes and underline important words.*
- *Students read one definition to their partner, and summarize in their own words what it is saying. Then, alternate who reads on each definition.*

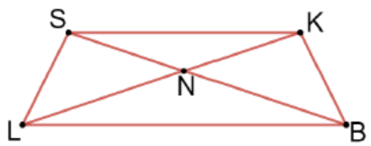
8.5.2 Collaboration: Properties of Isosceles Trapezoids

Component Overview: *In this collaborative activity students will measure segment lengths and angles of two isosceles trapezoids to determine if a statement is true or false. If not true, students will provide a conjecture that would be true.*

8.5.2 Collaboration: Properties of Isosceles Trapezoids

1. $RWDF$ and $SKBL$ are both ***isosceles trapezoids***. $\overline{RD}$ and $\overline{WF}$ are the diagonals of $RWDF$ and intersect at point Y. $\overline{SB}$ and $\overline{KL}$ are the diagonals of $SKBL$ and intersect at point N.





   a. Using a ruler and a protractor to measure $RWDF$ and $SKBL$, work with your partner to determine if the following conjectures about isosceles trapezoids are true or false. If false, write a new conjecture based on your measurements.

Encourage students to measure and organize their measurements prior to moving on to the table. Encourage students to choose whether they will split the work of measuring, or both measure everything and then compare. Allowing students to develop their own method of splitting the work teaches valuable cooperative and collaborative skills that can be used in the real world.

8.5.2 Collaboration: Properties of Isosceles Trapezoids (continued)

Conjecture	True or False?	If false, write a new conjecture.
All angles of an isosceles trapezoid are congruent	*False*	*Isosceles trapezoids have 2 pairs of congruent angles.*
Opposite sides are parallel	*False*	*One set of opposite sides are parallel.*
Opposites angles are congruent	*False*	*Opposite angles of an isosceles trapezoid are supplementary.*
Opposite sides are congruent	*False*	*One pair of opposite sides of an isosceles trapezoid are congruent.*
Diagonals bisect each other	*False*	*Diagonals of an isosceles trapezoid are congruent but do not bisect each other.*

b. Compare your chart to that of another pair. Discuss any differences. Make corrections as necessary.
Answers will vary.

Students sometimes choose to pick the answer that requires the less work, not because it is the correct answer. To avoid this, for the conjectures that students mark as true, set the expectation that they must present a drawing that supports the truth of the conjecture.

2. $\angle RFD$ and $\angle WDF$ in $RWDF$ are called ***base angles***. $\angle SLB$ and $\angle KBL$ are base angles in $SKBL$.

   Based on your previous measurements, make a conjecture about the base angles of an isosceles trapezoid.

   Base angles are congruent.

3. Each isosceles trapezoid has two pairs of base angles.
   a. What is the other pair of base angles in trapezoid $RWDF$?

      $\angle FRW$ and $\angle DWR$ are the other base angles.

   b. What is the other pair of base angles in trapezoid $SKBL$?

      $\angle LSK$ and $\angle BKS$ are the other base angles.

MA.K12.MTR.4.1: Discussion and Reflection
Students are encouraged to discuss and reflect with another pair of partners. Students should be encouraged to take risks and offer statements that others may disagree with. It is important as a class to meaningfully consider all opinions.

8.5.2 Collaboration: Properties of Isosceles Trapezoids (continued)

A ***trapezoid*** is a quadrilateral with *at least* one pair of parallel sides. It's a member of the following shape classes: polygons, quadrilaterals. Its sub-classes include parallelograms, rectangles, rhombuses, and squares.

An ***isosceles trapezoid*** is a trapezoid with two pairs of congruent base angles.

4. Make a conjecture about the legs of an isosceles trapezoid.

   Answers will vary. The legs of an isosceles trapezoid will be congruent.

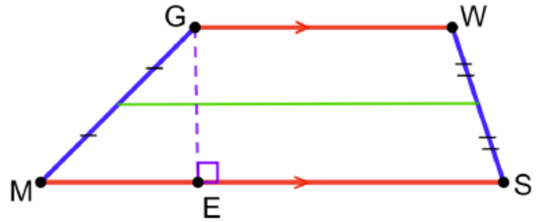
If an isosceles trapezoid has at least one pair of congruent base angles, what is true of the side lengths of an isosceles trapezoid?
Answer: You can use AAS to prove that opposite sides are congruent.

8.5.3 Exploration: Parts of a Trapezoid

Component Overview: *Students mark and explore the parts of a trapezoid to determine information that is true for a trapezoid.*

8.5.3 Exploration: Parts of a Trapezoid

1. Trapezoid $GWSM$ is shown.



a. The parallel sides of a trapezoid are called the ***bases***. The trapezoid shown has one pair of bases.

Label each base red and mark them to show they are parallel on $GWSM$.
The bases have been marked on the trapezoid in red.

b. The non-parallel sides of a trapezoid are called the ***legs***. The trapezoid shown has two legs.

Label each leg of $GWSM$.
The legs have been marked on the trapezoid in blue.

The **median** of a trapezoid connects the non-parallel sides at their midpoints. The median is sometimes referred to as the **midsegment**.

c. Using your ruler, plot the midpoint of $\overline{GM}$ and the midpoint of $\overline{WS}$. Connect these points to create the median of trapezoid $GWSM$ in green. Be sure to mark the legs appropriately showing that the median is connected at the midpoints.

The midpoints have been connected with a green line.

Consider creating an anchor chart with the parts of a trapezoid, and at the conclusion of 8.5.3 Exploration, add any pertinent information learned from question 2.

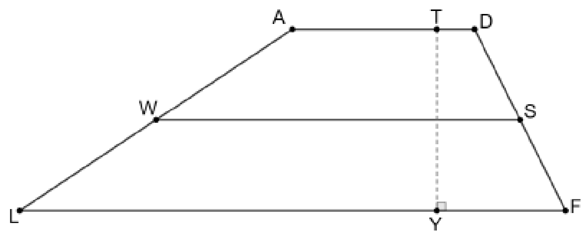
An **altitude** of a trapezoid is any segment from a point on one of the bases perpendicular to the line containing the other base.

d. The length of the altitude is the distance between the bases.

Draw the altitude on $GWSM$. Name the altitude by naming the endpoints and mark the trapezoid to show that the altitude is perpendicular to the base.

The altitude has been indicated on the trapezoid with a dashed purple line and is labeled $\overline{GE}$.

2. Trapezoid $ADFL$ is shown with midsegment $\overline{WS}$ and altitude $\overline{TY}$.



Select all of the information that is true about $ADFL$.

☐ $\overline{DF} \cong \overline{AL}$	☐ $m\angle LAD = m\angle FDA$	☐ $\overline{TY} \perp \overline{WS}$
☐ $\overline{AD} \parallel \overline{FL}$	☐ $\overline{WS} \parallel \overline{FL}$	☐ $m\angle AWS = m\angle DSW$
☐ $\overline{TY} \perp \overline{AD}$	☐ $\overline{DS} \cong \overline{SF}$	☐ $m\angle SFL = m\angle DSW$

This would be a great opportunity to do a "Live poll" using any "Voting" tool you like. Offer each option as a separate question for students to respond Yes, No, or I do not know. This data can be used as a formative assessment, if you count the number of each response for each question.

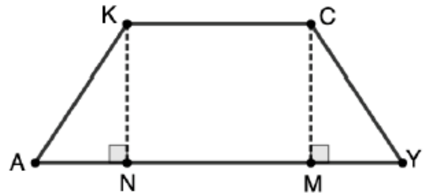
- *Count the number of incorrect or "I do not know" answers, and decide if this is too many to move on.*
- *Consider adding other possible statements to the poll to get a feel for the class.*

8.5.4 Guided Instruction: Proving Isosceles Trapezoid Properties

Component Overview: *Students will use patty paper to determine properties and develop theorems of isosceles trapezoids. Then, using the properties, guide the students through a two-column and a paragraph proof that will prove the theorems.*

8.5.4 Guided Instruction: Proving Isosceles Trapezoid Properties

1. Trapezoid $KCYA$ is shown with altitudes $\overline{KN}$ and $\overline{CM}$.



a. Trace the trapezoid onto a piece of patty paper.

b. Fold the patty paper so that vertices K and C coincide.

c. Describe what else you notice when you fold the patty paper.
Folding the patty paper created two congruent figures.

d. Work with a partner to list all of the congruent parts of trapezoid $KCYA$.
$\overline{KA} \cong \overline{CY}, \angle KAN \cong \angle CYM, \angle AKC \cong \angle YCK$

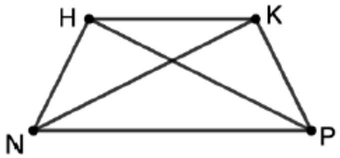
e. Compare your list with that of another pair. Add any congruent parts you may have missed.
Answers will vary.

f. Describe how the reflecting across the midpoint of $\overline{KC}$ shows that $KCYA$ can be classified as an isosceles trapezoid.
An isosceles trapezoid has congruent base angles. The reflection shows this trapezoid has two pairs of base angles congruent.

8.5.4 Guided Instruction: Proving Isosceles Trapezoid Properties (continued)

2. Given: $HKPN$ is an isosceles trapezoid with $\overline{HK} \parallel \overline{PN}$.

   Prove: $\overline{HP} \cong \overline{KN}$



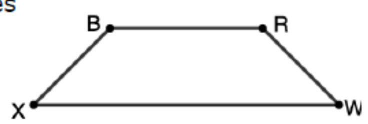
Complete the following two-column proof.

Statement	Reason
1. *$HKPN$ is an isosceles trapezoid; $\overline{HK} \parallel \overline{PN}$*	1. *Given*
2. $\overline{HN} \cong \overline{KP}$	2. *The legs of an isosceles trapezoid are congruent.*
3. $\angle HNP \cong \angle KPN$	3. Base angles of an isosceles trapezoid are congruent
4. $\overline{NP} \cong \overline{NP}$	4. *Reflexive Property of Congruence*
5. $\Delta HNP \cong \Delta KPN$	5. SAS Congruence Theorem
6. $\overline{HP} \cong \overline{KN}$	6. CPCTC

Consider implementing a silent board game for this proof. Display the proof and have students volunteer to go to the board and fill in one of the missing statements or reasons. There is no student talk during this activity. This routine offers students entry into the problem where they feel confident.

8.5.4 Guided Instruction: Proving Isosceles Trapezoid Properties (continued)

3. Given: $BRWX$ is an isosceles trapezoid and $\overline{BR} \parallel \overline{XW}$.
   Prove: $\angle RBX$ and $\angle RWX$ are supplementary, $\angle BRW$ and $\angle BXW$ are supplementary.



To add a challenge, provide the question stem, but not the partially completed proof. Have students complete the proof using any form they choose (two-column, paragraph, or flow-proof).

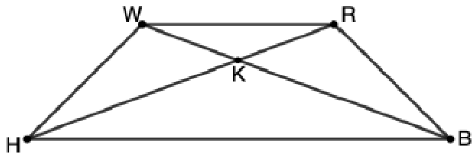
Given that $BRWX$ is an isosceles trapezoid, then $\angle RBX \cong \angle BRW$ because ___base angles of an isosceles trapezoid are congruent___. Therefore, by the definition of congruence, $m\angle RBX = m\angle BRW$. Since it is given that $\overline{BR} \parallel \overline{XW}$, then $\angle RBX$ and $\angle BXW$ are ___supplementary___ and $\angle$ RWX______ and $\angle$ BRW______ are supplementary because when parallel lines are intersected by a transversal, _consecutive interior angles_ are supplementary. Therefore, by definition of supplementary angles, $m\angle BRW + m\angle RWX =$ _180°_____ and _$m\angle RBX + m\angle BXW$_ $= 180°$. By substitution, $m\angle RBX + m\angle RWX = 180°$ and $m\angle BRW + m\angle BXW$ $= 180°$. Therefore, by _definition of supplementary angles__, $\angle RBX$ and $\angle RWX$ are supplementary and $\angle BRW$ and $\angle BXW$ are supplementary.

8.5.5 Guided Practice: Isosceles Trapezoids

Component Overview: *Students analyze a trapezoid and use properties of parallel lines, triangles, and other shapes to find missing angles and solve for unknown variable values.*

8.5.5 Guided Practice: Isosceles Trapezoids

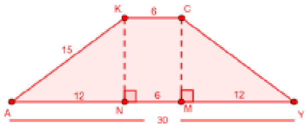
1. $WRBH$ is an isosceles trapezoid.



   a. If diagonal $WB = 18$ and diagonal $RH = 3x - 7$, find the value of x.
   $WB = RH$
   $18 = 3x - 7$
   $x = 8\frac{1}{3}$

   b. Diagonals $\overline{WB}$ and $\overline{RH}$ intersect at point K. If $m\angle RWB = 20.6°$, $m\angle RHW = 24.4°$, and $m\angle BKH = 138.8°$, what is $m\angle WRB$?
   $m\angle WKH = 180° - m\angle BKH = 180° - 138.8° = 41.2°$
   $m\angle HWK = 180° - m\angle RHW - m\angle WKH$
   $= 180° - 24.4° - 41.2° = 114.4°$
   $m\angle HWR = m\angle HWK + m\angle BWR = 114.4° + 20.6° = 135°$
   $m\angle WRB = m\angle HWR = 135°$

2. Isosceles trapezoid $KCYA$ has altitudes $\overline{KN}$ and $\overline{CM}$. If $KC = 6$, $AY = 30$, and $KA = 15$, what is the length of altitude $\overline{CM}$?
   The information is sketched and segment addition is used to determine the values of the segments that make up $\overline{AY}$.



   Use the Pythagorean theorem to solve for the altitude KN.
   $15^2 = 12^2 + (KN)^2$
   $KN = 9$
   $KN = CM = 9$

For students who are struggling, consider providing additional scaffolding for question 1 part b:

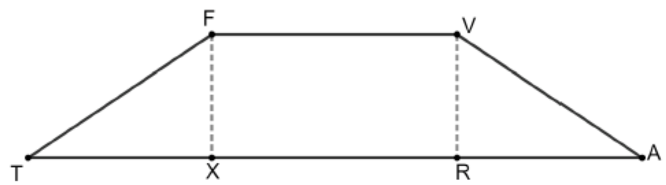
- *If the $m\angle BKH = 38.9°$, then $m\angle WKH =$ ______°* *because the two angles are ________________.*
- *$m\angle HWK =$ _______° because all three angles of a triangle equal _______°.*
- *$m\angle HWR = m\angle HWK + m\angle BWR =$ _______°*
- *$m\angle HWR = m\angle WRB =$ _______° because the __________ angles of isosceles triangles are congruent and equal.*

8.5.6 Wrap-Up: Ticket Out the Door

Component Overview: *Students use properties of trapezoids to find an unknown length. Students should work alone to demonstrate their knowledge of the properties of trapezoids. Students should not use a calculator to complete the Wrap-Up.*

8.5.6 Wrap-Up: Ticket Out the Door

1. Isosceles trapezoid $FVAT$ is shown with altitudes $\overline{FX}$ and $\overline{VR}$.



If $FT = 12$, $TA = 36\sqrt{5}$, and $VR = 8$, what is the length of $\overline{FV}$?

$FX = VR = 8$

$RA = TX$

Using the Pythagorean Theorem

$$TX = \sqrt{TF^2 - FX^2} = \sqrt{144 - 64} = 4\sqrt{5}$$

$TA = TX + XR + RA$

$36\sqrt{5} = 4\sqrt{5} + XR + 4\sqrt{5}$

$XR = 28\sqrt{5}$

$FV = XR = 28\sqrt{5}$

Use the data from 8.5.6 Wrap-Up as a formative assessment. This data can be used to determine students' mastery of the content presented in the lesson, and give information on which, and where, students might need further remediation.